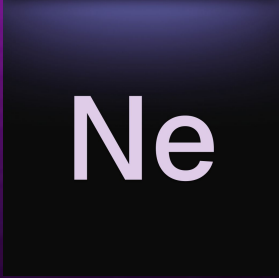


Low Level C++ Theory & Applications

Amlal El Mahrouss

I'm Amlal! Nice to meet you all!

The logo consists of a dark blue square with a subtle gradient and a slight shadow. Inside the square, the letters 'Ne' are written in a white, bold, sans-serif font.

Ne

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Purpose.

→ In this talk we're going to layout techniques and decisions to take in order to successfully develop a low-level system in modern C++.

The standards involved:

→ C++17 and later.

I: First Principles.

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Observation: What we will cover.

- Why C++ over C? (In N% percent of times)
- How C++ evolved (in a good way).
- What to learn from C++ in your codebase.

II: Common RAII.

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Observation: construct, use, destruct:

Use RAII for your containers:

→ `std::unique_ptr`.

→ `std::shared_ptr`.

→ `std::weak_ptr`.

→ `std::unique_lock`

RAII matters for resource sharing and management. One mismanaged resource can jeopardize the system.

III: RAII Patterns.

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IV: Applications and Examples of RAII.

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V: Final Remarks.

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VI: Conclusion.

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